

Project Documentation: Wi-Fi Security Auditor

1. Overview

The Wi-Fi Security Auditor is designed for Windows environments to perform configuration-based security assessments. By identifying the authentication and encryption protocols of visible Access Points (APs), the tool allows administrators to map wireless risk landscapes.

2. Wireless Security Concepts

To understand why this audit is necessary, it is vital to understand how wireless frames facilitate network connections. [Image of 802.11 wireless frame structure]
The tool specifically analyzes the management configuration of these frames to determine security posture.

3. Security Methodology

The tool uses an attribute-based detection logic to categorize network safety levels:

Protocol Detected	Security Status
Open / No Encryption	INSECURE
WEP	INSECURE
WPA2-PSK/Enterprise	SECURE
WPA3	SECURE

4. Technical Workflow

Execution Pipeline:

1. **Query:** The script invokes the Windows WLAN system via netsh.
2. **Extraction:** Regular expression patterns filter the structured output strings.
3. **Validation:** The logic flag verifies if detected protocols meet modern security baselines.
4. **Reporting:** Structured data is exported as a CSV file for long-term audit trail storage.

5. Ethical & Compliance

This audit tool should be integrated into standard infrastructure assessment workflows. Always ensure that the collection of wireless configuration data complies with internal company privacy and security policies.